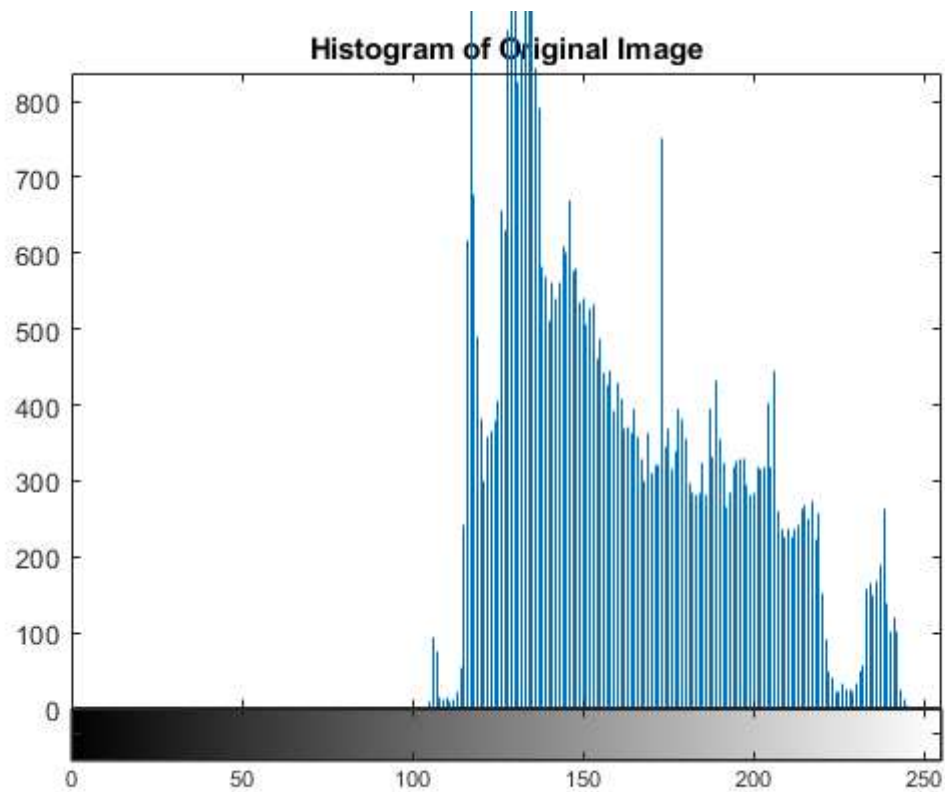


Problem 1

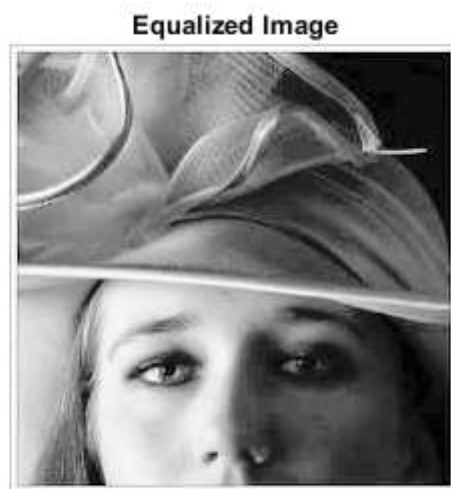
```
lowcontrast = imread("lowcontrast.jpg");  
lowcontrastgray = im2gray(lowcontrast);  
  
figure;  
imshow(lowcontrastgray)  
title('Original lowcontrast.jpg');
```



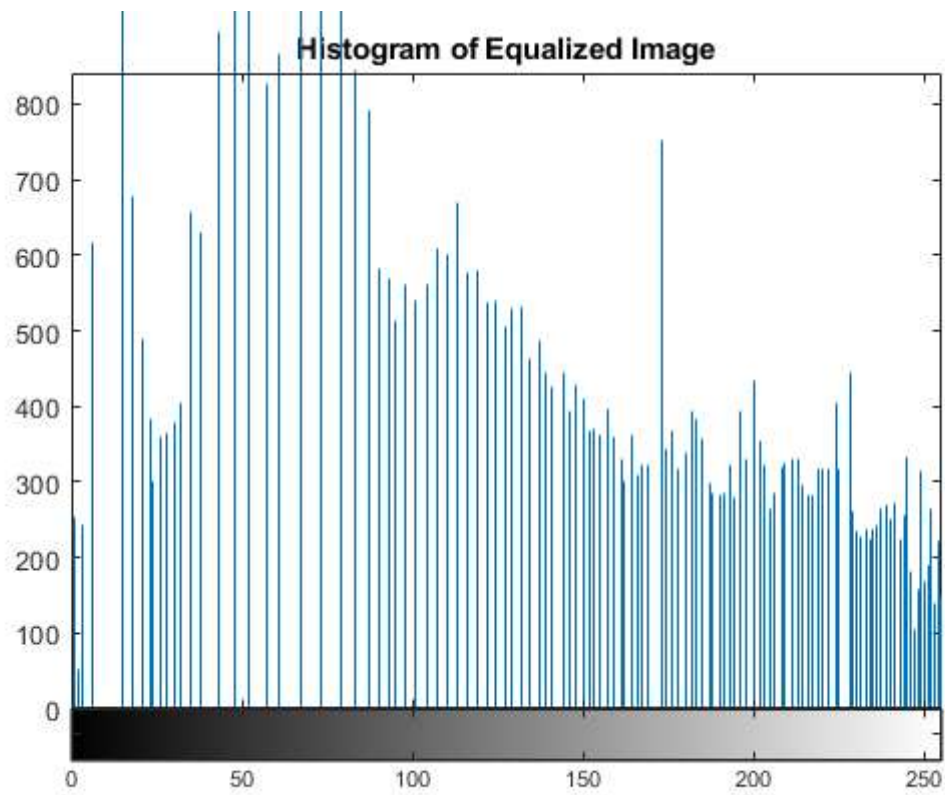
```
figure;  
imhist(lowcontrastgray)  
title('Histogram of Original Image')
```



```
figure;  
equalized = my_histeq(lowcontrastgray);  
imshow(equalized)  
title('Equalized Image')
```



```
figure;  
imhist(equalized)  
title('Histogram of Equalized Image')
```



Problem 2

```
lena_rgb = imread('lena.bmp');  
figure;  
imshow(lena_rgb)  
title('Original lena.bmp');
```

Original lena.bmp



```
lena_gray = rgb2gray(lena_rgb);  
  
a = edge(lena_gray, 'roberts');  
figure;  
imshow(a)  
title('Roberts Edge Detection');
```

Roberts Edge Detection



```
b = edge(lena_gray, 'prewitt');  
figure;  
imshow(b)  
title('Prewitt Edge Detection');
```

Prewitt Edge Detection



```
c = edge(lena_gray, 'sobel');  
figure;  
imshow(c)  
title('Sobel Edge Detection');
```

Sobel Edge Detection



Problem 3

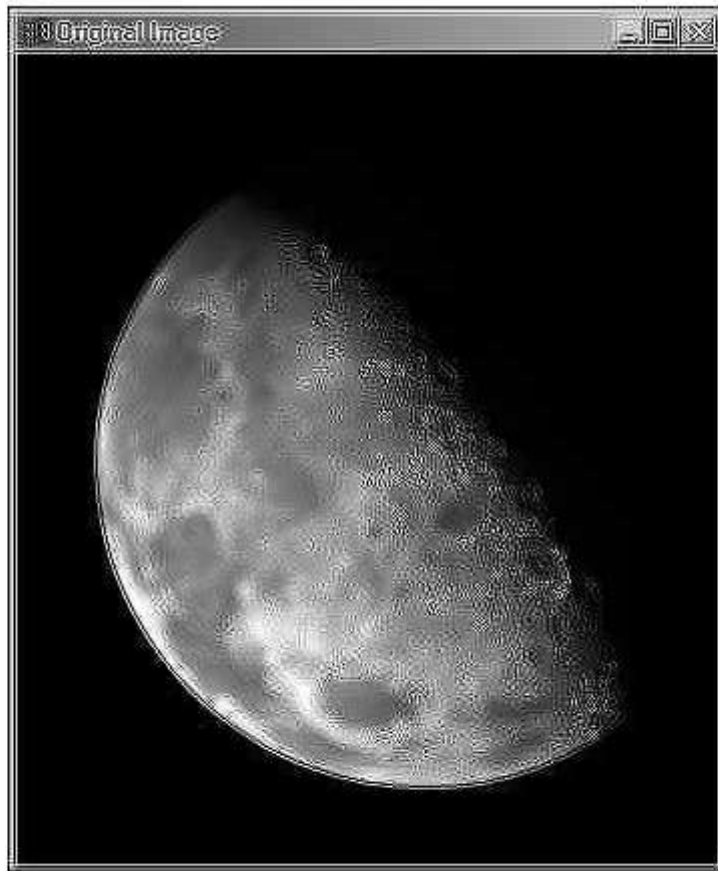
```
blur = imread('blur_image.jpg');  
  
figure;  
imshow(blur)  
title('Original blur_image.jpg')
```

Original blur_image.jpg



```
figure;  
img = laplacian(blur);  
imshow(img)  
title('Laplacian Sharpened blur_image.jpg')
```


Laplacian Sharpened blur_image.jpg



Functions

```
function eq = my_histeq(img)
    M = 255; % max intensity value
    counts = imhist(img); % get the original histogram
    pr = counts / numel(img); % PDF of original intensities
    cdf = cumsum(pr); % cdf = total probability
    mapping = M * cdf; % scale cdf by max intensity
    eq = uint8(mapping(double(img) + 1)); % apply mapping to each pixel
end

function blurred = laplacian(img)
    img = im2gray(img);
    img = double(img);

    L = [0, 1, 0;
         1, -4, 1;
         0, 1, 0];

    laplace = imfilter(img, L);
    blurred = img + laplace;
    blurred = uint8(blurred);
end
```

